



JS Capstone Project 1 - Students Version

 Title

EduTrack - A Student Progress Tracking - SaaS Application



Short Brief

You will work collaboratively as a team of 5 members for 4 days, utilizing the agile method with these roles (Scrum master, product owner, QA and developer; all team members are responsible for implementing the website)..

You will have a trainer follow-up to validate each step of the following during the 5 days of working on the project.

EduTrack is a browser-based **Software as a Service (SaaS)** platform that allows instructors, tutors, and training centers to track students' learning progress, record grades, provide feedback, and generate progress reports — all online.

The app will simulate a multi-user environment, where each instructor can manage their class data securely in the browser.

Tech Competencies Acquired

By completing this project, learners will gain:

- **JavaScript Mastery** – DOM manipulation, ES6+ syntax, event handling, and modular code.
 - **Data Structures** – Arrays, objects, and nested data for storing student records.
 - **State Persistence** – Using `localStorage` or IndexedDB to store user-specific data.
 - **API Simulation** – Mock or real API calls for fetching and storing progress reports.
 - **Dynamic UI** – Generating and updating student tables and charts in real time.
 - **Authentication Logic** – Basic sign-in/sign-out simulation for multiple instructors.
 - **SaaS Thinking** – Designing features to serve multiple users with isolated data.
-

Project Context

Modern education is shifting towards **cloud-based learning platforms**, and instructors need **real-time, accessible tools** to track performance.

EduTrack will mimic this environment, giving students a taste of building a **multi-user SaaS** while solving a real-world problem.

The app will:

- Let instructors **log in** and see their own dashboard.
- Add, edit, and delete student records.
- Track assignments, quizzes, and exam scores.

- Auto-generate visual charts for performance analytics.
-

Project Requirements

The application must include:

1. **Instructor Login (Simulated)**

- Each instructor logs in with their name/email.
- Only their own students' data appears.

2. **Student Management**

- Add a new student with name, ID, and enrolled courses.
- Update student scores and attendance records.
- Delete or archive student profiles.

3. **Progress Reports**

- Calculate total grades automatically.
- Display analytics with charts (e.g., grade distribution).

4. **Data Persistence**

- Store instructor and student data in `localStorage`.

5. **Responsive Layout**

- Optimized for mobile, tablet, and desktop views.
-

Performance Criteria

The project will be assessed on:

- **Functionality** – All listed features work without errors.
 - **Code Quality** – ES6+ features, modular functions, clear variable naming.
 - **UI/UX** – Easy to navigate, clean visuals, responsive design.
 - **SaaS Simulation** – Correctly isolates each instructor's data.
 - **Innovation** – Extra features like export to CSV, student search, or notifications.
-

Assessment Methods

- **Live Demo** – Walkthrough from login to report generation.
- **Code Review** – Instructors review and evaluate logic, structure, and readability.
- **Peer Testing** – Other students test and give feedback on usability.

Assessment Overview

Assessors should evaluate EduTrack for correctness, clarity, educational value, and production readiness within the scope of a client-side SaaS MVP. Total points = **100**.

Rubric (Each row scored 1–4)

Criterion (weight %)	Excellent (4)	Good (3)	Fair (2)	Poor (1)
Functionality (25%)	All required features implemented and work reliably (login simulation, add/edit/delete students, score calc, analytics).	Most major features implemented; small bugs.	Core features exist but broken in places.	Many missing features or non-working functionality.
Data Isolation & Persistence (15%)	Instructor data is fully isolated; localStorage schema stable; save/load works flawlessly.	Data mostly isolated; minor persistence edge cases.	Data mixes between users; persistence flaky.	No proper persistence or user isolation.
Analytics & Reports (10%)	Accurate grade calculations + clear charts/reports;	Reports mostly accurate; minor UI issues.	Reports exist but inaccurate or limited.	No analytics or broken reporting.

Criterion (weight %)	Excellent (4)	Good (3)	Fair (2)	Poor (1)
	export feature works.			
Code Quality & Structure (15%)	Modular, ES6+, well commented, reusable functions, sensible file structure.	Mostly modular; some code smell or missing comments.	Monolithic files; poor naming; few comments.	Hard to read, unmaintainable code.
UI/UX & Responsiveness (10%)	Clean, accessible UI; mobile-first and keyboard friendly.	Good layout; some responsiveness issues.	Barely responsive; usability problems.	Non-responsive, cluttered UI.
SaaS Simulation & Multi-User Logic (8%)	Login flow properly simulates different instructors; data scoped per user.	Works for basic flows but some edge cases.	Login simulation exists but data not fully scoped.	No user separation or broken flows.
Error Handling & Validation (7%)	Strong validation, graceful error messages, and input sanitization.	Basic validation present; some missing cases.	Minimal validation; visible JS errors.	No validation; crashes or silent failures.
Documentation & Deliverables (5%)	Complete README, setup, screenshots, API notes, and demo video.	README present; missing some details.	Minimal README, no screenshots.	No documentation.
Innovation / Bonus Features (5%)	Extra features (CSV export, filters, history) well-implemented.	One or two extras implemented.	Attempted extras but incomplete.	No extras.



Grading Scheme & Passing Thresholds

- Max score = **100**.
- Passing threshold = **70 / 100**.
- Grade bands:
 - **85–100** = Excellent (A)
 - **70–84** = Satisfactory (B)
 - **50–69** = Needs Improvement (C) — remediation required
 - **<50** = Fail (D/F) — major rework required

Weights from rubric above apply to compute final numeric grade.



Deliverables

- Fully working **JavaScript SaaS application** (EduTrack).
- Source code folder with HTML, CSS, JavaScript, and assets.
- README file with setup steps, feature list, and screenshots.
- Optional short video walkthrough of features.